

Max Ratcliff

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Education

University of California, Santa Cruz, Santa Cruz, CA

Expected June 2027

B.S Double Major: Applied Mathematics & Computer Engineering (Systems Concentration)

3.6/4.0

Relevant Coursework:

- **Systems & Arch:** Logic Design (Verilog), Computer Networking, Computer System Design, Assembly & Architecture, Computer Systems & C, Operating Systems.
- **Math & ML:** Scientific ML, AI Research, ML & Data Mining, Linear Algebra, ODEs/PDEs, Vector Calculus, Probability/Statistics.

Skills

Programming: Python, C, C++, Verilog, RISC-V ASM, MATLAB, Java, JavaScript, Fortran 90, HTML/CSS, LaTeX

Libraries & Frameworks: PyTorch, JAX, NumPy, SciPy, Pandas, Matplotlib, FastAPI, React, Node.js, Docker, CellPose

Tools & Platforms: Git, Linux (Kernel/System Admin), Google Cloud, Firebase, Supabase, MS Office, Adobe CC

Languages: English (Native), German (Native)

Experience

Software Engineering Intern, AMI

April 2026 — Present

- Engineering a centralized LLM orchestration layer utilizing the **LiteLLM** abstraction framework to provide a unified OpenAI-compatible API for disparate providers (Gemini, Claude, GPT-4), ensuring seamless model swapping without code changes.
- Architecting an internal **Prompt Management System** featuring version-controlled templates and metadata-driven injections; developed RESTful endpoints to decouple prompt engineering from application logic, reducing deployment cycles for LLM-driven features.
- Implementing comprehensive **usage guardrails** and commercial key isolation to prevent PII leakage and enforce cost-tracking; designed middleware for token-count validation and per-team rate limiting to maintain budgetary control.
- Authoring exhaustive API documentation and technical specifications for the prompt management lifecycle, including schema definitions for prompt variables, model-specific parameters, and fallback logic for high-availability systems.

Undergraduate Researcher (Neuromorphic Computing), UC Santa Cruz

Jan 2026 — April 2026

- Investigated the alignment of **Spiking Neural Networks (SNNs)** with biological manifolds by adapting **Joint-Embedding Predictive Architectures (JEPA)**, focusing on how predictive objectives can enforce "brain-like" representations in sparse temporal data.
- Built training pipelines using **snntorch** (LIF neurons) and **LeJEPA** as a teacher model; validated alignment using **Canonical Correlation Analysis (CCA)** to quantify the similarity between artificial neural trajectories and recorded biological activity.
- Developed interpretable neural circuit motifs by filtering stochastic noise through JEPA's predictive objectives; researched how contrastive-free training prevents representation collapse in neuromorphic systems without requiring negative samples.
- Utilized **Google Colab A100 GPUs** for high-throughput temporal backpropagation through time (BPTT), optimizing SNN hyperparameters to match biological firing rates while maintaining competitive task accuracy.

SciAI Pathways Scholar, UC Santa Cruz (ARO Lab)

June 2024 — Present

- Developing a high-fidelity **multimodal wound healing model** in **PyTorch** that fuses visual features from **ResNet-50** embeddings with transcriptomic (RNA) genomic signatures processed via a Multi-Layer Perceptron (**MLP**) to predict patient outcomes.
- Implementing a late-fusion architecture to handle disparate data modalities, utilizing transfer learning on limited clinical datasets while exploring embedding-based alignment to prevent overfitting in the genomic branch.
- Mentoring first-year researchers on end-to-end ML workflows, including data cleaning for noisy microscopy sets, PyTorch Dataset/DataLoader optimization, and modular code design for collaborative research.
- Engineered a fully automated, headless data pipeline in **Python** using **CellPose** (custom-trained weights) to segment 5,000+ microscopy images, extracting morphological features (area, eccentricity) with **99% accuracy**.
- Optimized image preprocessing (CLAHE, Gaussian blurring) and batch processing to achieve **0.3ms latency** per image, enabling real-time spore population feedback for closed-loop bioreactor control.

Research Assistant (AI Curriculum), UC Santa Cruz

Feb 2025 — Aug 2025

- Designed and implemented an AI-driven calculus curriculum for **500+ students**, integrating LLMs as personalized tutoring assistants to address foundational knowledge gaps.
- Analyzed student sentiment and performance metrics, identifying a divergent impact: increased conceptual proficiency but a decline in manual computational skills among heavy AI users.
- Contributed to a research publication (in progress) and provided data-driven recommendations for university-wide AI policy and integration.

Founding Developer, Slug AI / PieFi (Consulting Group)

Jan 2025 — Feb 2025

- Recruited as one of 3 founding engineers for **PieFi**, an automated VC platform; built a lead discovery dashboard using the **CLADO semantic people search API**.
- Developed an automated cold-outreach system that generated personalized emails based on semantic analysis of lead profiles and project details.
- Previously served as a founding member of Slug AI, a student-run consulting group focused on rapid AI/ML prototyping for local startups.

Resident Assistant, UC Santa Cruz

Sept 2024 — Present

- Direct leadership resource for **60 residents** and 24/7 on-call emergency lead for a community of **2,500+ students**; managing conflict resolution and crisis intervention.
- Mentor new RAs on effective conflict resolution and community building strategies; lead community-building programs for diverse student populations.

Senior Lifeguard & Instructor (Supervisor), City of Pasadena

Summers 2020 — 2025

- Acted as third-in-command for daily pool operations, supervising staff and managing safety for **300+ daily swimmers**.
- Instructed 2,000+ students in Red Cross curriculum, teaching water safety and swimming skills at all levels.
- Promoted from Junior Lifeguard to Senior Supervisor due to excellence in crisis management, team coordination, and facility safety standards.

Projects

Benefit Finder

Jan 2025 — Present

- Developed a **TypeScript** and **Next.js** platform at **CruzHacks 2025** that utilizes **Gemini Deep Research APIs** to parse complex government eligibility PDFs and match them against user-provided demographic JSON blobs.
- Designed a privacy-first data handling layer that processes sensitive user information locally where possible, and implemented a PDF form auto-fill engine to automate the application lifecycle for users.
- Collaborating with the **City of Santa Cruz** to pilot the tool, focusing on SOC2-compliant data handling and creating a modular "benefit scraper" that can be updated via LLM-driven schema mapping.

Calendarize — AI Media-to-Calendar Tool

calendarize.ratcliff.cc

- Developed a full-stack **FastAPI** and **React** application using the **Gemini-1.5-Flash** API to perform Vision-to-JSON extraction from unstructured media (screenshots, Instagram flyers).
- Implemented the **Google Calendar OAuth2** flow and designed a background worker system to handle event creation retries and time-zone normalization for over **1,000 active users**.
- Deployed the application using a **Dockerized** multi-stage build on **Google Cloud Run**, leveraging auto-scaling to handle traffic spikes during university event seasons.

Autoregressive Weather Forecasting (VAE)

ratcliff.cc/projects/weathermodel

- Architected a **convolutional Variational Autoencoder (VAE)** in PyTorch to compress 2D atmospheric data into a 128-dimensional latent space, achieving **0.00032 MSE** in reconstructing missing sensor data.
- Implemented a **conditional autoregressive** decoder that predicts the next 12-hour state by sampling from the latent distribution, effectively modeling temporal dependencies without the vanishing gradient issues of standard RNNs.
- Validated model robustness against "mode collapse" by visualizing the KL-divergence during training and performing recursive 48-hour forecasting, maintaining physical consistency in temperature and pressure gradients.

Hybrid Neural Network Chess Engine

chess.ratcliff.cc

- Engineered a hybrid chess engine pairing a **Minimax search with Alpha-Beta pruning** and a **dual-head ResNet** (Policy head for move ordering, Value head for position evaluation).
- Trained the ResNet on **160 Million positions** from the Lichess Elite database (2500+ ELO) using **A100 GPUs**; implemented a bitboard representation and custom move generator to maximize training throughput.
- Optimized inference via **policy-based move ordering**, reducing the search tree branching factor and allowing the engine to reach 6-ply depth in $\leq 100\text{ms}$ (**15,000 nodes/sec**) on consumer hardware.
- Achieved a stable **1647 ELO** in self-play matches against Stockfish Level 5, demonstrating superior tactical awareness in complex mid-game transitions.